

CS 2005: ARTIFICIAL INTELLIGENCE

Project Requirement Description

Full Project (end of Week 15)

Total Marks: 10

AI Project – Technical Requirements

1) DESIGN PHASE:

a) Project Idea

- i. Introduction and Description: Briefly describe the problem your system will solve.
- ii. System Functionality: Explain what your system will do.
- iii. Users of the System: Identify users and their roles.
- iv. Technologies Used: Mention tools/technologies for implementation.

Important Requirements:

- The project must be based on AI techniques.
- Students may use any programming language.
- UI (Graphical Interface) is mandatory.
- Visual representation of algorithm is compulsory.

b) Algorithm Selection

- i. Select at least one technique from:
 - BFS, DFS, A*
 - Local Search
 - Genetic Algorithm (GA)
 - CSP (Constraint Satisfaction Problem)
 - Bayesian Network
- ii. Clearly explain the working of the selected technique.

2) IMPLEMENTATION PHASE:

Implement the system using your selected technologies.

Your implementation must include:

- Working UI
- Algorithm Logic
- Visual Representation

Visualization Requirements:

a) Search Algorithms:

- Show node exploration and final path

b) CSP:

- Show variables, domains, constraints
- Show backtracking and assignments

c) Genetic Algorithm:

- Show population and evolution

d) Bayesian Network:

- Show nodes, edges, and probability updates
- Show inference based on user input

Additional Requirements:

- Code must be clean and well-structured
- Proper comments must be included
- Project should be meaningful and not too basic

SUBMISSION INSTRUCTIONS

- Submit complete project in one ZIP file
- Include report, code, and screenshots
- Submit on Google Classroom
- Only group leader will submit
- Mention all group members with IDs